# 12.9 A Fair Decision from a Biased Coin

WORKSHEET

**The question**

**STEP I 2019, Question 11**

Read the question before you go on. It is in the STEP I paper in OCR's download of the 2019 STEP papers (a zip file):

www.ocr.org.uk/Images/696590-2019-past-papers.zip

## Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

**Ask yourself** In one round, what is the probability of a decision? Of no decision? What is the largest value $2pq$ can take?

**STAGE 2 Decision by round $n$**

**Ask yourself** What is the probability of *no* decision in the first $n$ rounds?

**STAGE 3 Three people**

**Ask yourself** After a first round with two heads and one tail, the third person turns the tail over to a head. What must happen in round two for a decision? And after two tails and one head?

**Ask yourself** Write everything in terms of $s = pq$. (Use $p + q = 1$.)

## Try this next

1. **Expected rounds.** Find the expected number of rounds for the procedure in part (i), and for von Neumann's procedure. For which $p$ is von Neumann's procedure slower?

2. **Fair three-way choice.** Using a biased coin, devise a procedure that chooses one of three options, each with probability exactly $\frac{1}{3}$.

3. **A biased die from a fair coin.** Using a fair coin, devise a procedure that simulates a coin with probability of heads exactly $\frac{1}{3}$. What is the expected number of tosses?

4. **Three people again.** In part (ii), what is the probability that they go to the hotel, as a function of $p$? Is the procedure fair when $p = \frac{1}{2}$?

## Further reading


- **John von Neumann, "Various techniques used in connection with random digits"**, *National Bureau of Standards Applied Mathematics Series* 12 (1951). Short and famous.
- **Donald Knuth, *The Art of Computer Programming*, Vol. 2** (Addison-Wesley), Chapter 3, on random numbers.
- **Michael Mitzenmacher & Eli Upfal, *Probability and Computing*** (Cambridge University

Press), on extracting randomness and entropy.

The full guided solution, and what the question leads to at university, is in Section 12.9:
https://beyondstep.keithmcnulty.org/ch12/von-neumann.html